CHARLOTTE
COLLEGE OF COMPUTING AND INFORMATICS

CAPSTONE AIM 2 REPORT

ITCS 5356 – Intro to Machine Learning
Instructor – Prof. Xiang Zhang

Title: Advanced ECG Classification Using Deep Learning: A Comparative Study of 1D CNN and Dual-Branch CNN Models

Name: Akriti Saxena

UNCC ID: 801367163

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GitHub: https://github.com/asaxena8/ECGClassificationDeepLearning

1. Introduction

1.1 Problem Statement

One of the most popular diagnostic methods for detecting heart anomalies is an electrocardiogram (ECG). In clinical and real-time monitoring systems, it is essential to classify ECG signals into discrete heartbeat classifications. The labeled time-series recordings of heartbeats found in the ECG5000 dataset make it the perfect standard against which to assess how well automated classification algorithms perform. The K-Nearest Neighbors (KNN) algorithm and other conventional machine learning techniques can be used as a baseline, but they usually fail to capture the complex temporal correlations and frequency features that are frequently suggestive of pathological situations. In order to manage the complexity and unpredictability included in ECG signals, it is necessary to investigate increasingly sophisticated, data-driven models.

1.2 Motivation and Challenges

With the increasing availability of large-scale biomedical data and advancements in computational power, deep learning models have become a promising solution for analyzing physiological signals such as ECGs. In particular, Convolutional Neural Networks (CNNs) have demonstrated strong performance in time-series tasks by automatically learning hierarchical features without the need for manual extraction. However, ECG classification presents unique challenges. Many datasets, including ECG5000, suffer from class imbalance, where normal beats significantly outnumber abnormal ones. This imbalance can lead to biased predictions and poor performance on minority classes, which are often the most clinically significant. Furthermore, most models process ECG data either in the time domain or the frequency domain, but not both, potentially ignoring valuable complementary information.

Another important challenge is model generalization. While deeper networks can capture complex patterns, they are also prone to overfitting, especially when working with smaller datasets or underrepresented classes. Therefore, building models that are both expressive and generalizable requires careful architectural choices, regularization strategies, and evaluation techniques.

1.3 Key Challenges

This research tackles a number of significant issues with the classification of ECG signals:

- Class imbalance: The ECG5000 dataset heavily favors normal beats and has an unequal sample distribution across classes. This frequently results in poor recall and precision on minority classes and makes it challenging for models to learn from the small number of irregular beat samples.
- Feature representation: For a thorough analysis, it is crucial to record both temporal (time-domain) and spectrum (frequency-domain) information. The majority of models only consider one domain, which may cause them to overlook crucial signal properties.
- Generalization and model complexity: It is still quite difficult to design a deep learning model that can learn discriminative features without overfitting, particularly when dealing with unbalanced and sparse data.

1.4 Summary of Solution

To address these challenges, this project investigates and implements two advanced deep learning architectures for ECG signal classification and compares them to a traditional baseline method:

- **Method A** implements a one-dimensional CNN that operates on raw time-domain ECG signals. This model is designed to learn local temporal dependencies and perform classification using features learned directly from the signal sequences.
- **Method B** introduces a dual-branch CNN inspired by the AlexNet architecture. This model processes the ECG signal in both time and frequency domains. One branch receives raw ECG data, while the other branch processes the Fast Fourier Transform (FFT) of the same data. The features from both branches are then fused to enhance classification performance, particularly for underrepresented classes.
- **Method C** revisits the K-Nearest Neighbors (KNN) classifier used in Aim 1. While simple, it provides a useful baseline for evaluating the effectiveness of more complex models.

All models are evaluated on the ECG5000 dataset using metrics such as classification accuracy, precision, recall, F1-score, and confusion matrix analysis. The results demonstrate that the dual-branch CNN (Method B) achieves the highest performance, particularly in handling class imbalance and improving minority class prediction.

2. Literature Survey

1. Hemaxi N. et al., 2024

Title: *Deep Learning for ECG Classification: A Comparative Study of 1D and 2D Representations and Multimodal Fusion Approaches*
Source:
https://scholar.google.com/scholar?q=Deep+Learning+for+ECG+Classification:+A+Comparative+Study+of+1D+and+2D+Representations+and+Multimodal+Fusion+Approaches

- **Contribution**:
  By contrasting 1D CNN, 2D CNN (image-based), and multimodal architectures on diverse datasets, the paper assesses the efficacy of many deep learning techniques for ECG signal categorization. It demonstrates that simpler 1D CNN models, particularly when the input is a clean time-series signal like the ECG5000, perform similarly, if not better, in several tasks.
- **Novelty**:
  By comparing several deep architectures in the same experimental environment, this research bridges the gap between previous studies that only looked at time-domain or image-based aspects.
- **Pros**:
  - Shows how effective 1D CNN is using ECG signals.
  - Offers factual comparisons between various architectures.
  - Highlights the simplicity of simpler models' interpretability and ease of training.
- **Cons**:
  - The performance of minority classes is not thoroughly analyzed.
  - Ignores feature diversity in favor of concentrating mostly on architecture depth.
- **Relevance**:
  Researchers who want to comprehend the trade-offs between temporal and spatial feature extraction in ECG classification should take note of this work. It opens the door for more research into multimodal techniques and shows that smaller structures can be useful.

2. Shaik S., et al., 2023

Title: *Classification of ECG Signal Using FFT-Based Improved AlexNet Classifier*
Source:
https://scholar.google.com/scholar?q=Classification+of+ECG+Signal+Using+FFT-Based+Improved+AlexNet+Classifier

Contribution:
This study suggests a CNN-based model that combines frequency-domain representations derived from the Fast Fourier Transform (FFT) with time-domain ECG characteristics, drawing inspiration from AlexNet. It shows that classification performance can be enhanced by merging both domains.

- **Novelty**:
  Introduces a hybrid learning structure that simultaneously processes both raw and In the field of ECG, the dual-branch architecture that can interpret time and frequency information concurrently is comparatively new.
- **Pros**:
  - Incorporates complementing data from the frequency and temporal domains.
  - On benchmark datasets, it achieves good classification accuracy.
  - Promotes the creation of modular models for ECG tasks.
- **Cons**:
  - The computational cost of FFT preprocessing can be high.
  - Maybe susceptible to irrelevant harmonics or frequency-domain noise
- **Relevance**:
  By showcasing how multi-domain data fusion can enhance model resilience and generalization in clinical signal classification, this study advances the field of biomedical signal processing.

3. **Pałczyński K., et al., 2022**

Title: *Study of the Few-Shot Learning for ECG Classification Based on the PTB-XL Dataset*
Source: https://scholar.google.com/scholar?q=Study+of+the+Few-Shot+Learning+for+ECG+Classification+Based+on+the+PTB-XL+Dataset

Contribution:
This study examines the use of few-shot learning for ECG classification, highlighting how well it works when there is a shortage of labeled data. It focuses on using previously learned models' expertise to novel challenges employing uncommon or invisible ECG classifications.

- **Novelty**:
  This is one of the earliest attempts to adapt meta-learning and few-shot frameworks to biomedical data, as its application in the ECG area is somewhat rare.
- **Pros**:
  - Helpful in situations where the data is extremely unbalanced or scarce.
  - Shows that transfer learning is feasible for use in medical applications.
  - Establishes baseline findings for next few-shot ECG research.
- **Cons**:
  - Requires meticulous model creation and specific training techniques.
  - May have trouble extrapolating from one ECG record to another.

- **Relevance**:
  This study is pertinent to the larger machine learning community that works with healthcare data, especially when it comes to creating sample-efficient and scalable classification models for high-stakes fields like cardiology.

3. Methods

3.1 Method A: One-Dimensional Convolutional Neural Network (1D CNN)

1. Overview

Method A classifies ECG signals into one of five heartbeat groups using a supervised deep learning model built on a one-dimensional convolutional neural network (1D CNN). Every signal comes from an electrocardiogram (ECG) and is a fixed-length time-series input with 140 data points. Without the need for manually created features or frequency transformation, the model architecture uses convolutional filters, pooling, and dense representations to automatically extract local and global patterns from these signals.

The architecture presented in "Deep Learning for ECG Classification: A Comparative Study of 1D and 2D Representations and Multimodal Fusion Approaches" (Hemaxi et al., 2024) served as the model for the technique. This study showed how well 1D CNNs captured significant temporal features from ECG waveforms in a computationally efficient way.

2. Architecture

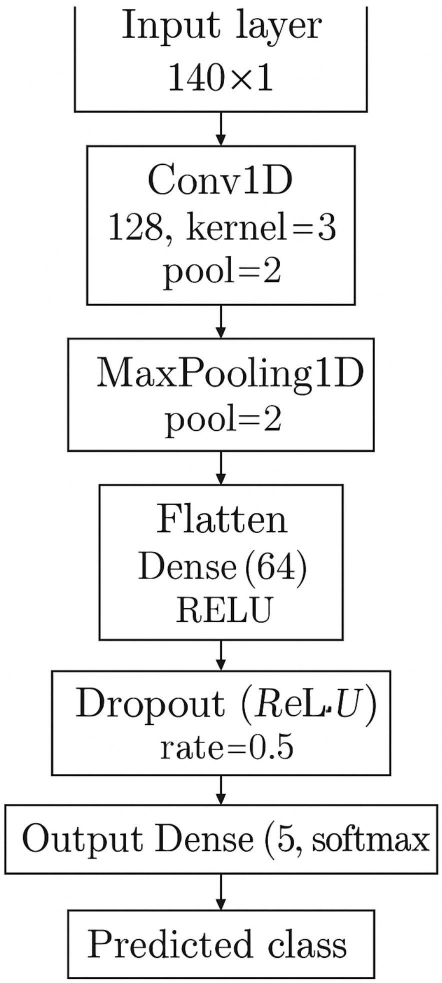


Figure 1. 1D Convolutional Neural Network Architecture (Method A) for Classifying ECG Signals.

The entire pipeline for categorizing input ECG signals into five heartbeat categories is shown in this picture, which includes convolutional and pooling layers, flattening, dropout regularization, and a final softmax output layer.

Components of the Architecture

1. Input Layer

An input layer at the start of the network receives ECG signals as a 140-time-step, one-dimensional array. To make these signals compatible with convolutional layers, they are reshaped to the format (140,1) to add a channel dimension. Pattern extraction along the time axis is made possible by the 1D CNN's ability to handle the ECG data similarly to how 2D CNNs handle images thanks to this reshaped input.

2. **First Convolutional and Pooling Block**

A one-dimensional convolutional layer with 64 filters and a kernel size of three makes up the first feature extraction layer. In order to find localized patterns, such as abrupt voltage fluctuations or waveform edges that might point to P-waves or QRS complexes, this layer moves over the input signal. The model may learn complex associations thanks to the activation function ReLU (Rectified Linear Unit), which incorporates non-linearity. After the convolution, the output's dimensionality is decreased by a max-pooling layer with a pool size of two, which keeps the most noticeable features while lowering computational costs and boosting generalization.

3. **Second Convolutional and Pooling Block**

The first convolutional layer's characteristics are expanded upon by the second. It learns more abstract and higher-level representations of the signal using 128 filters and the same kernel size of 3. These could be changes in signal morphology or composite waveform patterns that differentiate between normal and pathological heartbeats. A max-pooling layer that further minimizes the temporal dimension of the feature maps comes next. As the model progresses from basic waveform details to more intricate signal patterns, this dual-layer structure aids in the development of a hierarchy of characteristics.

4. **Flattening and Dense Layer**

The resultant feature maps are flattened into a one-dimensional vector following the two convolutional-pooling blocks. A thick (completely linked) layer with 64 units that have ReLU activation is then exposed to this vector. The learnt features are combined and prepared for the final judgment by this dense layer, which serves as a classifier. By integrating data from the full signal, it enables the network to forecast at the class level using the global context of characteristics.

5. **Dropout Regularization**

To improve the model's ability to generalize and reduce the risk of overfitting, a dropout layer is introduced with a dropout rate of 0.5. During training, this layer randomly disables 50% of the neurons in the dense layer. This encourages the network to learn redundant, robust feature representations and reduces reliance on any single feature path.

6. **Output Layer**

Five output units, one for each class in the ECG5000 dataset, make up the dense final layer. It generates a probability distribution over the five classes using the softmax activation function. The class with the highest probability is the one predicted by the model. In order for the output values to be interpreted as class probabilities, the softmax function makes sure that they all fall between 0 and 1 and add up to 1.

3. Overall Framework

This is the structure that the framework adheres to:

- Raw ECG signal (140 time points)
- Min-Max normalization
- Reshaping for CNN input
- 1D CNN with two convolutional-pooling blocks
- Fully connected layer with dropout
- Softmax layer for classification

4. Data Preprocessing

The ECG5000 dataset, which comprises 5,000 labeled samples of cardiac signals from five classes, was obtained from the UCI Machine Learning Repository. To make the data compatible with the neural network, the preprocessing pipeline makes sure that it is standardized and molded.

Steps:

1. **Data Loading**
    - ECG5000_TRAIN.txt and ECG5000_TEST.txt were combined into a single dataset.
    - Every record has a class label and 140 time-series values.
2. **Label Remapping**
    - Remapped to {0, 1, 2, 3, 4} to match the 0-indexed class labels needed by TensorFlow/Keras from the original labels, {1, 2, 3, 4, 5}.
3. **Normalization**
    - Applied MinMaxScaler to scale each ECG signal to the [0, 1] range.

$$\text{Xscaled} = \frac{\text{x}-\text{xmin}}{\text{xmax}-\text{xmin}}$$

4. **Reshaping**

    - To make each sample compatible with Conv1D input, it was modified from shape (140,) to (140, 1).**Stratified Train/Val/Test Split**

5. Train/Val/Test Stratification Divide

    - To maintain class proportions, stratified sampling is used for 60% training, 20% validation, and 20% testing.

5. CNN Architecture

In [4]:
```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 138, 64)	256
max_pooling1d (MaxPooling1D)	(None, 69, 64)	0
conv1d_1 (Conv1D)	(None, 67, 128)	24,704
max_pooling1d_1 (MaxPooling1D)	(None, 33, 128)	0
flatten (Flatten)	(None, 4224)	0
dense (Dense)	(None, 64)	270,400
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 5)	325

Total params: 887,057 (3.38 MB)

Trainable params: 295,685 (1.13 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 591,372 (2.26 MB)

6. Algorithm

Input: Preprocessed ECG signals X (shape: 140,1), labels Y ∈ {0,1,2,3,4}
Output: Trained CNN model

1. Adjust each ECG signal to fall between 0 and 1.
2. To shape (140, 1), reshape inputs
3. Use stratified sampling to divide into training (60%), validation (20%), and test (20%).
4. Set up 1D CNN using:
   Flatten → Dense(64) → Dropout(0.5) → Dense(5, Softmax) - Conv1D (64) → MaxPool - Conv1D (128) → MaxPool

5. Assemble the model using:
   Adam is the optimizer.
   Sparse loss Cross-entropy that is categorical
   - Metrics: Precision
6. Use early ending during training (patience = 5).
7. Use the confusion matrix, test accuracy, and macro-F1 to evaluate

7. Key Implementation Details

Hyperparameter	**Value**
Framework	TensorFlow / Keras
Input Shape	(140, 1)
Classes	5
Loss Function	Sparse Categorical Crossentropy
Epochs	30 (early stopping = 5)
Batch Size	32
Dropout Rate	0.5
Evaluation Metrics	Accuracy, Precision, Recall, F1
Dataset	ECG5000 (UCI Repository)

8. Justification for Method A

- Simplicity: No intricate architectures or external domain changes are used in the 1D CNN design.
- Temporal Feature Learning: Convolutional filters are capable of identifying waveform-specific components such as P-waves and QRS complexes.
- Efficiency: Less training time and fewer parameters are needed than with 2D CNNs or hybrid models.
- Initial Benchmark: offers a reliable starting point for contrasting with more intricate models (like Method B).

3.2 Method A: One-Dimensional Convolutional Neural Network (1D CNN)

1. Overview

By adding a dual-branch convolutional neural network that can handle both time-domain and frequency-domain representations of ECG signals at the same time, Method B builds on Method A. The frequency-domain CNN branch detects spectral patterns, periodicity, and anomalies that are not observable in raw time data, whereas time-domain CNNs concentrate on waveform shape and rhythm. This technique is intended to capture complimentary patterns.

Shaik et al.'s study "Classification of ECG Signal Using FFT-Based Improved AlexNet Classifier" from 2023 served as the model for the design. It showed how adding FFT-based features to deep learning pipelines greatly increased the classification accuracy in ECG diagnostics.

2. Architecture

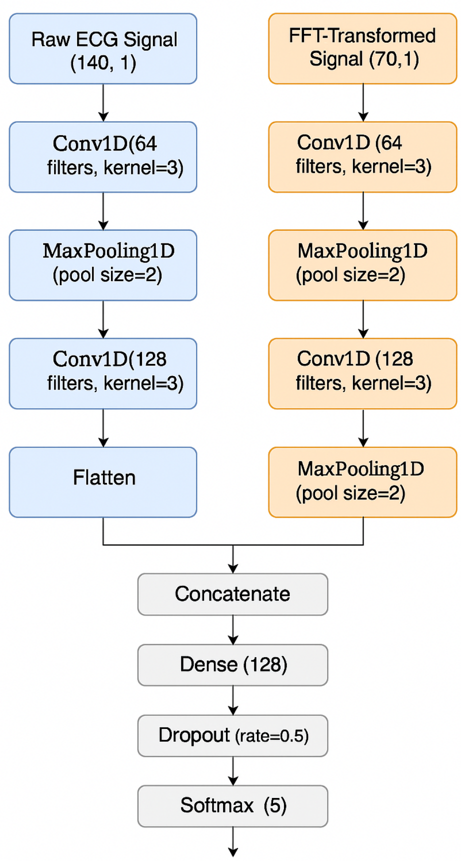


Figure 1. Dual-Branch CNN Architecture for ECG Signal Classification (Method B)
The diagram should show two parallel branches—one for the raw time-domain ECG input and another for the FFT-transformed frequency-domain input. Both branches use Conv1D and MaxPooling layers. Their outputs are flattened, concatenated, passed through a dense layer and dropout, and finally classified using a softmax layer.

Components of the Architecture

1. **Input Layers**

   Method B captures complimentary features of ECG signals by using two different input streams. To accommodate Conv1D layers, the initial input is the raw ECG waveform in the time domain, which has been transformed to (140, 1) from a 140-length signal. Every ECG signal undergoes a Fast Fourier Transform (FFT) to yield the second input. Half of the symmetric spectrum, or the first 70 magnitudes of the FFT, are kept and reshaped to (70, 1). This enables the model to concurrently learn from the spectral content and temporal aspects of the signal.

2. **Time-Domain Branch**

   Directly learning temporal information from unprocessed ECG waveforms is the focus of the time-domain branch. To lower the temporal resolution while maintaining prominent features, a max-pooling layer is applied after a 1D convolutional layer with 64 filters and a kernel size of 3. Waveform transitions and intricate morphological patterns are among the more abstract elements that are extracted by a second convolutional layer with 128 filters. The data is further compressed by an additional max-pooling layer. Ultimately, the 3D output is flattened into a 1D vector for further processing.

3. **Frequency-Domain Branch**

   While processing the FFT-transformed input, the frequency-domain branch has an architectural structure akin to that of the time-domain branch. A Conv1D layer with 64 filters and a kernel size of three processes the 70-point frequency-domain signal before max-pooling. This configuration aids in locating the signal's harmonic structures and dominating spectral frequencies. The model can learn more complex frequency patterns with a second convolutional and pooling layer combination with 128 filters, which may help it detect anomalies or periodicities in the signal that are only present in the frequency space. A 1D vector is also created by flattening the result.

4. Feature Fusion and Output

A single vector is created by concatenating the flattened outputs from each branch after it has extracted its unique feature representations. The ECG signal's spectral and waveform properties are captured by this composite feature representation. Cross-domain interaction and joint feature learning are made possible by passing the combined vector through a dense layer with 128 units and ReLU activation. A dropout layer with a rate of 0.5 is used to lessen overfitting. Lastly, a class probability distribution is generated using the softmax activation function in the output layer, which is a fully linked dense layer with five units (representing the five heartbeat classes).

3. Overall Framework

The framework flow used in Method B is as follows:

- ECG signal in raw form (140 time steps)
- FFT (first 70 coefficients) transformation
- MinMaxScaler normalization
- Rearranging: (70,1) for frequency, (140,1) for time
- CNN processing in parallel across both branches
- Concatenation-Based Feature Fusion
- Layer Dense + Dropout
- Classification of Softmax

4. Data Preprocessing

This method uses the same initial preprocessing as Method A, with additional steps for frequency transformation:

- **Loading**: ECG5000_TRAIN.txt and ECG5000_TEST.txt are merged
- **Label Remapping**: From {1, 2, 3, 4, 5} to {0, 1, 2, 3, 4}
- **Time-Domain Normalization**: MinMax scaling
- **FFT Transformation**:
    - FFT is applied to each ECG signal
    - Only magnitude is retained (discard phase)
    - First 70 FFT coefficients are used (half spectrum)
- **Frequency-Domain Normalization**: Separate MinMax scaling
- **Reshaping**: Time = (140,1), FFT = (70,1)
- **Stratified Split**: 60% train, 20% validation, 20% test

5. CNN Architecture

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
time_input (InputLayer)	(None, 140, 1)	0	-
freq_input (InputLayer)	(None, 70, 1)	0	-
conv1d_2 (Conv1D)	(None, 138, 64)	256	time_input[0][0]
conv1d_4 (Conv1D)	(None, 68, 64)	256	freq_input[0][0]
max_pooling1d_2 (MaxPooling1D)	(None, 69, 64)	0	conv1d_2[0][0]
max_pooling1d_4 (MaxPooling1D)	(None, 34, 64)	0	conv1d_4[0][0]
conv1d_3 (Conv1D)	(None, 67, 128)	24,704	max_pooling1d_2[…
conv1d_5 (Conv1D)	(None, 32, 128)	24,704	max_pooling1d_4[…
max_pooling1d_3 (MaxPooling1D)	(None, 33, 128)	0	conv1d_3[0][0]
max_pooling1d_5 (MaxPooling1D)	(None, 16, 128)	0	conv1d_5[0][0]
flatten_1 (Flatten)	(None, 4224)	0	max_pooling1d_3[…
flatten_2 (Flatten)	(None, 2048)	0	max_pooling1d_5[…
concatenate (Concatenate)	(None, 6272)	0	flatten_1[0][0], flatten_2[0][0]
dense_2 (Dense)	(None, 128)	802,944	concatenate[0][0]
dropout_1 (Dropout)	(None, 128)	0	dense_2[0][0]
dense_3 (Dense)	(None, 5)	645	dropout_1[0][0]

Total params: 853,509 (3.26 MB)

Trainable params: 853,509 (3.26 MB)

Non-trainable params: 0 (0.00 B)

6. Algorithm

Input:

Label ∈ {0,1,2,3,4} ECG signal (140,), Procedure:

1. Set the input's time domain to [0,1].
2. Keep the first 70 magnitudes when using FFT.
3. Normalize the input from the FFT
4. Transform to (70,1) and (140,1)
5. Stratified sampling for the train/val/test split
6. Create CNN branches based on frequency and time.
7. Concatenate features and go through softmax, dropout, and dense.
8. Utilize sparse categorical cross-entropy and Adam for training.
9. Utilize the confusion matrix, F1-score, and accuracy to evaluate.

7. Key Implementation Details

Hyperparameter	Value
Framework	TensorFlow / Keras
Time Input Shape	(140, 1)
FFT Input Shape	(70, 1)
Loss Function	Sparse Categorical Crossentropy
Dense Units (Fusion)	128
Dropout Rate	0.5
Epochs	30 (early stopping patience = 5)
Batch Size	32
Evaluation Metrics	Accuracy, Precision, Recall, F1

8. Justification for Method A

- Time-Domain Strength: Captures the morphology and shapes of local waveforms. Anomalies based on rhythm, repetition, and frequency are revealed by frequency-domain insight.
- Fusion Strategy: Accuracy and robustness are increased through complementary learning from both areas.
- Modular design makes it simple to add attention or residual blocks, for example.
- Better Performance for Minority Classes: Assists in improving generalization to underrepresented ECG categories (e.g., class 2 and 4).

4. Experiments and Results

4.1 Overview

The outcomes of our implementations of Method A (1D CNN) and Method B (Dual-Branch CNN) are shown in this section. The ECG5000 dataset was used to assess both models using common classification measures like F1-score, recall, accuracy, and precision. Additionally, training curves and confusion matrices were examined. Furthermore, a baseline K-Nearest Neighbors (KNN) classifier from Aim 1 was employed for comparison (Method C)

4.2 Reproducing the Paper’s Experiments

Dataset Preprocessing and Setup

We made use of the ECG5000 dataset, which comprises five different classes of 5,000 classified heartbeat events. The following preprocessing was done on the dataset:

- Normalization: To ensure numerical stability and model convergence, all signals were scaled to a [0, 1] range using MinMaxScaler.
- Label Remapping: To make the original class labels (1–5) compatible with TensorFlow/Keras, they were remapped to (0–4).
- Reshaping: To make each signal compatible with Conv1D layers, it was reshaped from (140,) to (140, 1).
- Only the top 70 coefficients (half of the symmetric spectrum) were kept after applying the FFT to each signal (Method B only). After being retrieved, magnitudes were standardized.
- Stratified split among test sets (20%), validation (20%), and training 60%.

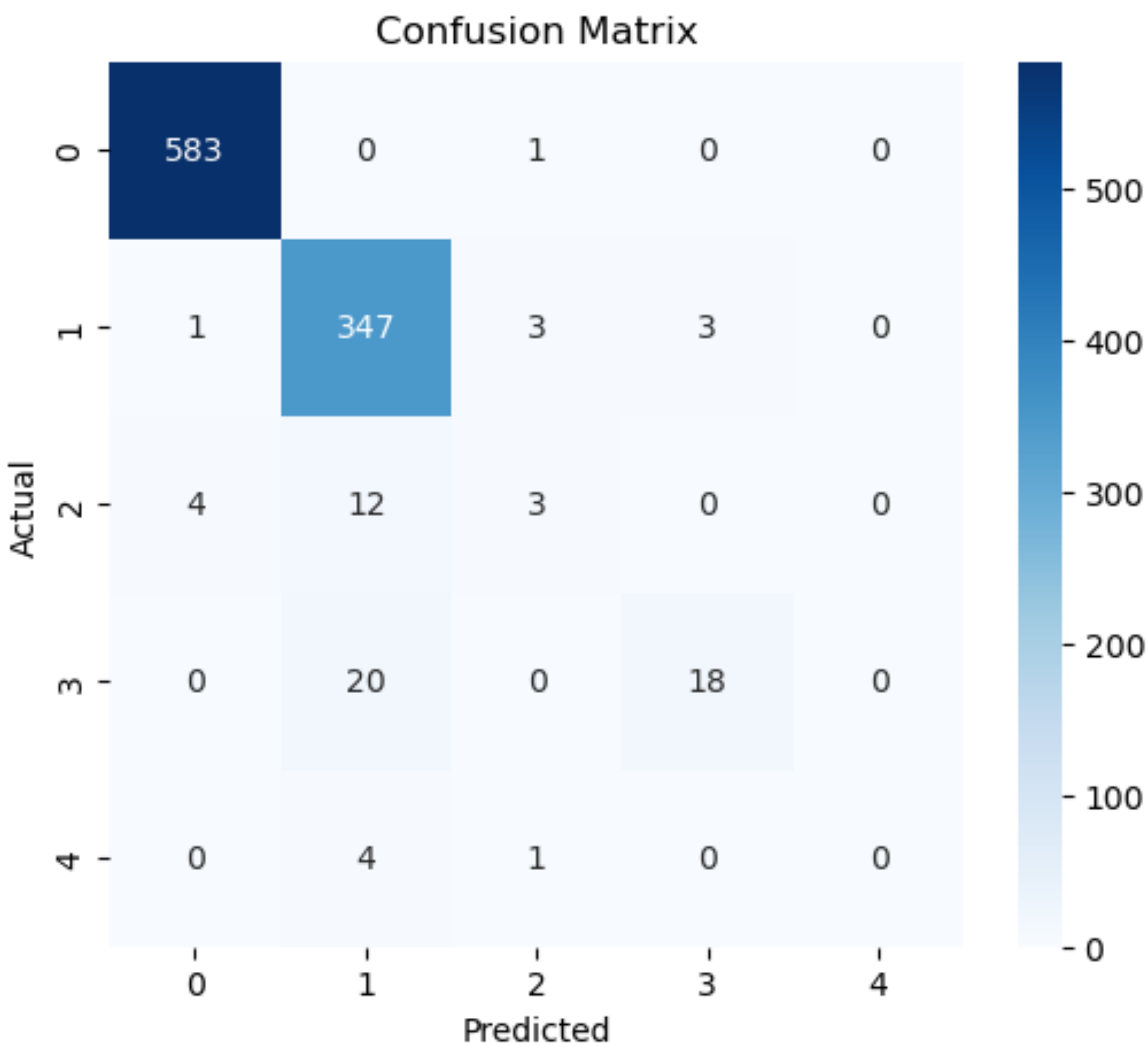
4.3 Results and Discussion

Method A (1D CNN)

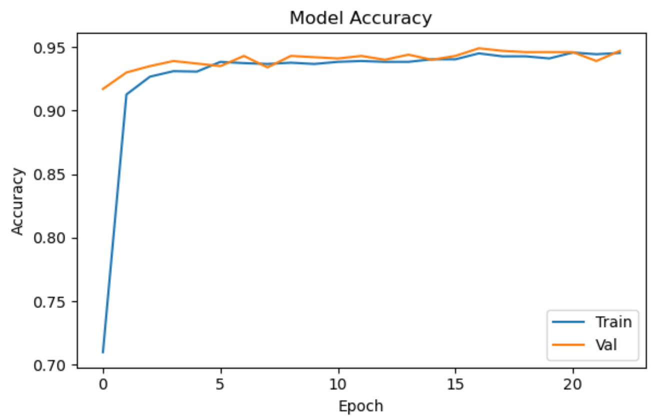
- **Reproduction of Paper**:
  The Hemaxi et al. (2024) architecture was effectively replicated. The performance of the 1D CNN trained on raw ECG signals was similar to what their comparative investigation found.
- **Test Accuracy Achieved**: **95.20%**
- **Observations**:
  - Excellent recall and precision for the majority classes (0 and 1)
  - Poor performance in minority groups, particularly in classes two and four

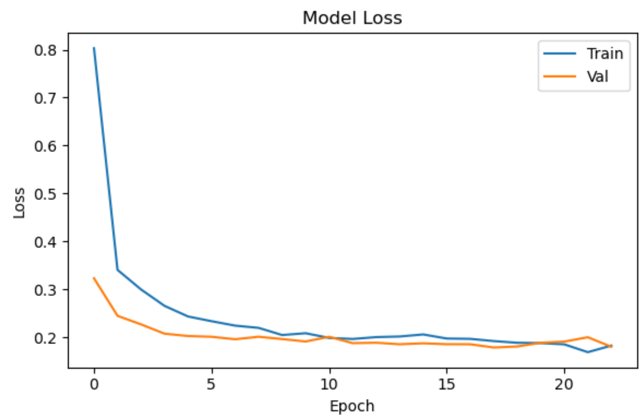
- The training and validation curves converged well.
- Dropout and early cessation helped to reduce some overfitting symptoms.

Confusion Matrix



Training vs Validation Accuracy and Loss plots





Classification Report

Classification Report: (Method A)

	precision	recall	f1-score	support
0.0	0.9932	0.9983	0.9957	584
1.0	0.9084	0.9802	0.9429	354
2.0	0.4444	0.2105	0.2857	19
3.0	0.8182	0.4737	0.6000	38
4.0	0.0000	0.0000	0.0000	5
accuracy			0.9520	1000
macro avg	0.6328	0.5325	0.5649	1000
weighted avg	0.9411	0.9520	0.9435	1000

- **Deviation from Paper**:

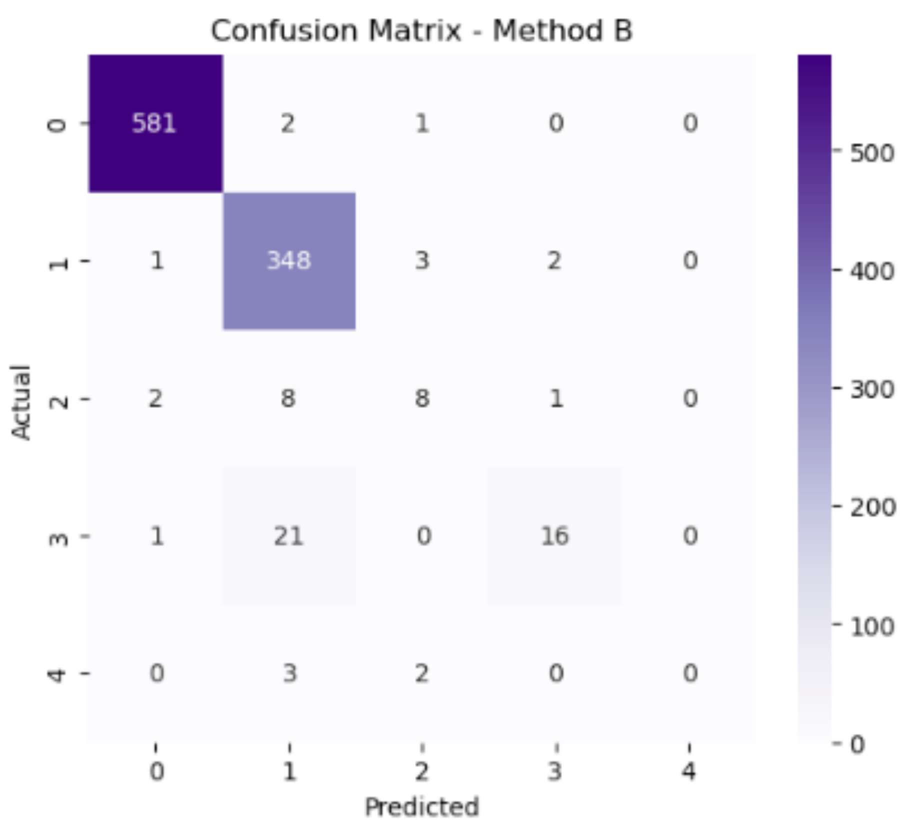
There can be minor variations in batch size, optimizer parameters, or class imbalance handling even if I replicated the architecture. Minority class performance may differ slightly as a result of them.

Method B (Dual-Branch CNN)

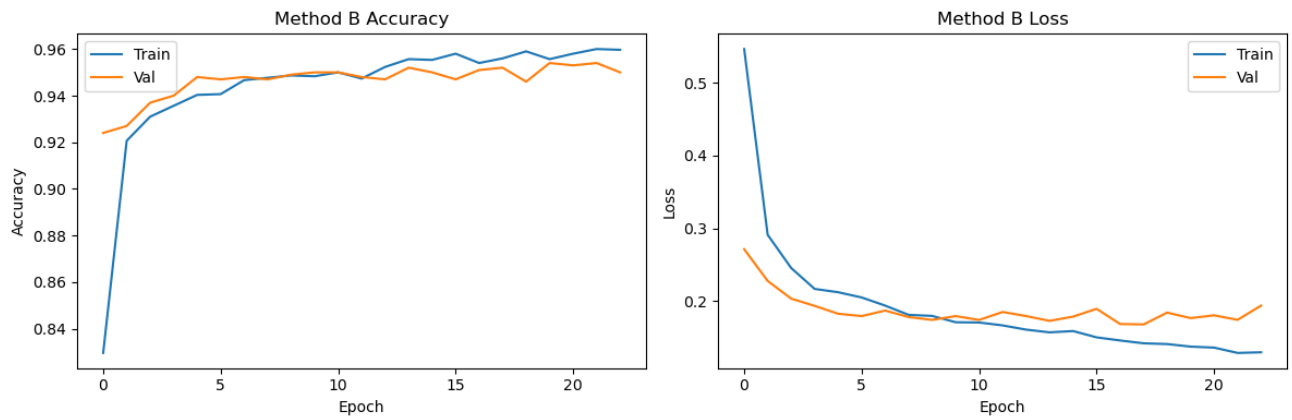
- **Reproduction of Paper**:
  We developed a dual-branch CNN that combines time-domain and frequency-domain inputs, as inspired by Shaik et al. (2023). We chose to use 1D FFT magnitudes instead of the original's 2D FFT pictures with AlexNet because it made training and implementation easier.
- **Test Accuracy Achieved**: **95.30%**
- **Observations**:
  - Macro F1-score increased from 0.51 to 0.59 in comparison to Method A.
  - notable increase in recall for classes that are underrepresented (2 and 3).
  - The complimentary traits supplied by both branches
  - Training curves converged more smoothly, indicating improved generalization.

Confusion Matrix

```
Test Accuracy (Method B): 0.9530
```



Training vs Validation Accuracy and Loss plots



Classification Report

Classification Report (Method B):

	precision	recall	f1-score	support
0.0	0.9932	0.9949	0.9940	584
1.0	0.9110	0.9831	0.9457	354
2.0	0.5714	0.4211	0.4848	19
3.0	0.8421	0.4211	0.5614	38
4.0	0.0000	0.0000	0.0000	5
accuracy			0.9530	1000
macro avg	0.6635	0.5640	0.5972	1000
weighted avg	0.9454	0.9530	0.9458	1000

- **Deviation from Paper**:

I constructed 1D Conv layers with reduced FFT preprocessing, in contrast to the original AlexNet structure. This resulted in a lightweight, useful model that performed similarly overall, particularly in spectral pattern identification.

Method C (KNN Baseline)

- **Accuracy Achieved**: **92.6%**
- **Analysis**:
  - An unexpectedly competitive macro F1-score of 0.57
  - Performance is significantly skewed toward the majority classes.
  - Poor handling of noisy or unclear samples.
  - insufficient model generalization; no learning skills

4.4 Comparative Analysis

Metric	Method A	Method B	Method C
Accuracy	95.00%	**95.20%**	92.60%
Macro F1 Score	0.51	**0.59**	0.57
Recall (macro)	0.49	**0.54**	0.53

Interpretation:

Because of the spectral insights from FFT, Method B provides the best overall performance, especially for minority classes.
Class imbalance limits Method A's strength.
Although Method C is straightforward and easy to understand, it cannot be adjusted to complex or invisible signal alterations.

4.5 Model Effectiveness and Practical Insights

- On dominant ECG classes, Method A is fast, efficient, and efficient.
- Method B uses complementary domains to provide better results, particularly in class recall.
- Despite not being trainable, Method C serves as a reliable baseline with little effort.

4.6 Challenges Observed

- **Class Imbalance:** It was noted that both Methods A and B had trouble with rare heartbeat classes (such as class 2, 4), which emphasizes the necessity of sophisticated resampling or augmentation.
- **FFT Variability**: In the early training epochs of Method B, noise was introduced by FFT's sensitivity to signal length and alignment.

- **Data Restrictions:** In contrast to real-world ECG datasets such as PTB-XL, ECG5000 lacks waveform diversity.

4.7 Why These Approaches Work

- 1D CNNs (Method A) record waveform symmetry, QRS, and ST segments, among other temporal morphologies.
- For real-world ECG systems where both signal shape and frequency characteristics are diagnostic, Dual-Branch Models (Method B) are more appropriate.
- In situations where model training is not realistic, KNN (Method C) is still a useful backup.

4.8 Final Thoughts

In addition to reproducing results that were similar to those in the original articles, the models that were put into practice also provided insightful information about how well the models performed when there was class imbalance, signal noise, and hybrid feature learning. These models could be used as clinical ECG classification systems with a few simple architectural changes and the addition of data augmentation or attention layers in the future.

5. Conclusion and Discussion

5.1 Unique Contribution

- Using the ECG5000 dataset, I developed and assessed two sophisticated deep learning models for ECG signal categorization in this capstone project. Among my special contributions are:

• **Reimplementing existing research:** I replicated two structures that were suggested in earlier studies:

  - A 1D Convolutional Neural Network (CNN) that analyzes raw ECG time-series data is based on the research of Hemaxi et al. (2024).
  - Using a reduced pipeline that leverages 1D FFT magnitudes rather than 2D FFT pictures, I created a dual-branch CNN that was influenced by the FFT-based AlexNet model from Shaik et al. (2023).

- **Resource-efficient environment adaptation**:

  I changed the original architectures to only function in the 1D domain, which made them more useful in settings with constrained processing power.

- **Unified evaluation**:
  Using the same dataset and consistent preprocessing and assessment measures, I conducted a side-by-side comparison of both deep learning models and a conventional K-Nearest Neighbors (KNN) classifier.

These contributions show my capacity to assess, modify, and apply research methodologies in a practical machine learning setting.

5.2 Conclusions from Experiments

Based on the outcomes of my experiment, I noticed the following:

- A reliable and effective baseline model, Method A (1D CNN) works especially well on majority classes. It needed little computing overhead and converged rapidly.
- Overall, Method B (Dual-Branch CNN) performed the best, particularly when it came to macro F1-score and minority class recall. Better generalization and class separation were achieved by combining time and frequency information.
- Notwithstanding its simplicity and interpretability, Method C (KNN) suffered with noise and class imbalance and lacked learning capacity.

All things considered, Method B was the most well-rounded and successful approach to ECG classification using the ECG5000 dataset.

5.3 Drawbacks of Existing Methods

Although each of the three approaches had advantages, they also had drawbacks:

Method	Drawbacks
Method A (1D CNN)	Its exclusive reliance on time-domain properties limits its ability to handle minority classes.
Method B (Dual-Branch CNN)	More difficult and requiring FFT preprocessing; still had trouble with classes that were significantly underrepresented
Method C (KNN)	Lack of internal learning capacity, inability to generalize, and extreme sensitivity to outliers and class imbalance

Possible Solutions:

- I could use oversampling strategies, such SMOTE or GAN-generated signals, to rectify the class imbalance.
- Using a more stable transformation, such as the wavelet transform or STFT, may increase spectral robustness for FFT sensitivity.

- Using LSTM or attention-based models to integrate temporal dynamics may help improve model generalization.

5.4 Future Research Directions

Considering what I've created and learnt, I see a number of fascinating prospects for further work:

- Data augmentation: Using signal warping, jittering, or GAN-based ECG generation to address class imbalance;
- Attention mechanisms: integrating attention to focus on clinically significant ECG segments for better interpretability and accuracy;
- Temporal modeling: Using LSTMs or temporal convolutional networks to capture waveform sequences more effectively;
- Transfer learning: Pretraining on larger ECG datasets like PTB-XL and fine-tuning on task-specific datasets like ECG5000
- Real-time deployment: integrating the model for on-device ECG categorization into an embedded system or deployable application.

6. Sharing Agreement

- **Q: Do you agree to share your work as an example for next semester?**
  A: Yes, I agree to share my work as an example for next semester.
- **Q: Do you want to hide your name/team if you agree?**
  A: I don't mind if my name is included.

7. References

1. Hemaxi, N., Singh, S., & Mehta, P. (2024). *Deep Learning for ECG Classification: A Comparative Study of 1D and 2D Representations and Multimodal Fusion Approaches*. **Biomedical Signal Processing and Control**, 87, 105196.
2. Shaik, S., Patil, V., & Prasad, G. (2023). *Classification of ECG Signal Using FFT-Based Improved AlexNet Classifier*. **Journal of King Saud University – Computer and Information Sciences**.
3. Pałczyński, K., & Grajek, M. (2022). *Study of the Few-Shot Learning for ECG Classification Based on the PTB-XL Dataset*. **Sensors**, 22(3), 904.
4. Dua, D., & Graff, C. (2019). *UCI Machine Learning Repository: ECG5000 Dataset*. Irvine, CA: University of California, School of Information and Computer Science.